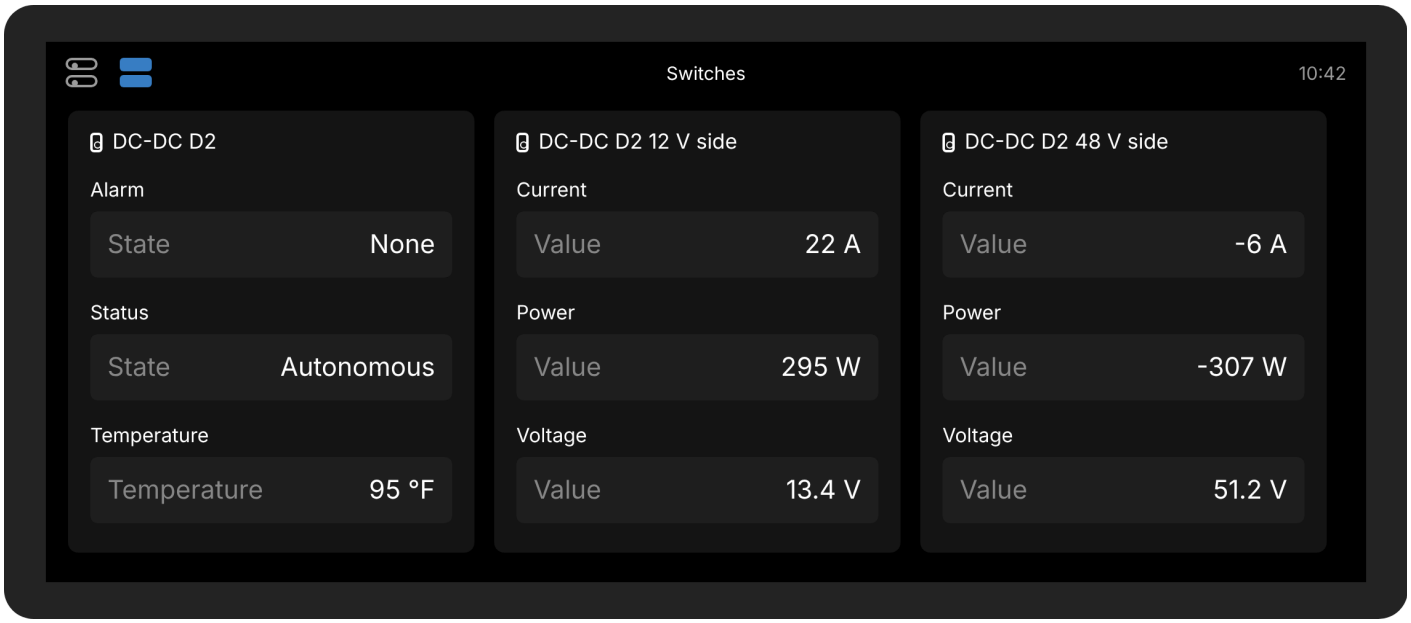
INSTALLATION GUIDE · MONITOR V2.1

Nomadic D2 DC-DC converter monitor on your Victron Cerbo GX

Make the cable, set up the Cerbo, and install the monitor. When you're done, the D2's readings appear on the Cerbo screen, in the VRM app and on a web page with live charts.

It only watches. The monitor listens to the D2 and never sends it anything, so the D2 keeps working exactly as it does today, on its own built-in logic.



What you'll see on the Cerbo screen when you're done (drawn from Victron's own screen design, so the real screen may differ slightly).

You'll be able to see

- 12 V and 48 V side voltage, current and power
- The D2's temperature, what it's doing (standby, autonomous, derating…) and every alarm, in plain words
- Charts of the last 24 hours on a web page
- Everything in the VRM app on your phone

Good to know

- It takes about 1 hour, including making the cable
- Nothing changes for the D2. The Cerbo can't send it any commands
- You don't need to type any commands. The **Cerbo Setup** helper does it for you

STEP 1 Get what you need

STEP 2 Make the cable

STEP 3 Set up the Cerbo

STEP 4 Connect everything

STEP 5 Install the monitor

STEP 6 Use it

1 Get what you need

Gather everything before you start.

For the cable

- A standard Ethernet patch cable (Cat5e or Cat6), long enough to reach from the Cerbo to the D2
- Two **3-way lever wire connectors**, for example WAGO 221-413. They take 24–12 AWG wire: solid-core Cat5e/Cat6 (about 24 AWG) fits; very thin stranded patch-cable wire may not clamp well, so use solid-core cable, or put a small ferrule on each end
- Wire cutters and a wire stripper
- Heat-shrink tubing or electrical tape
- A multimeter with a continuity (beep) setting and an Ohms setting
- A VE.Can terminator: the small RJ45 plug that came in the Cerbo box
- A **120 Ω resistor**, ¼ W metal film (1 %), for the D2 end. Measure it before use: 115–125 Ω

For the software

- A Windows PC on the same network (Wi-Fi or Ethernet) as the Cerbo
- The project folder downloaded from GitHub, which contains **Cerbo Setup.bat** and the **dbus-nomadic-dcdc** folder
- The Cerbo running Venus OS **v3.80 or newer**. Step 3 shows you how to check

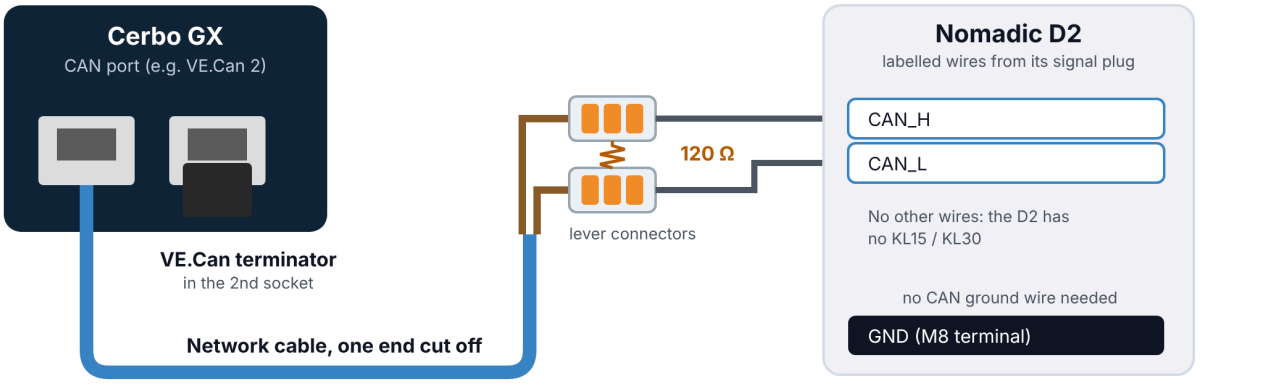
Safety

Before you work on the D2's wiring, open the fuses or breakers on **both** its LV+ (12 V) and HV+ (48 V) leads and follow the D2 manual. The D2 has no KL30/KL15 supply to switch off: it powers itself from whichever power terminal is live. The CAN cable only carries data, but it ends next to the D2's 48 V and 12 V power terminals.

2 Make the cable

The Cerbo uses **RJ45 sockets**, the same plug as network cables. Take a ready-made network cable, keep the plug on one end for the Cerbo, and cut off the other end. You only need **2 of its 8 wires**: the brown pair. The D2 already has labelled wires coming out of its signal plug (CAN_H and CAN_L). You join the network cable to its CAN_H and CAN_L wires with **two lever wire connectors**. The **120 Ω resistor** goes between the two connectors. **No ground wire** is needed.

The whole connection

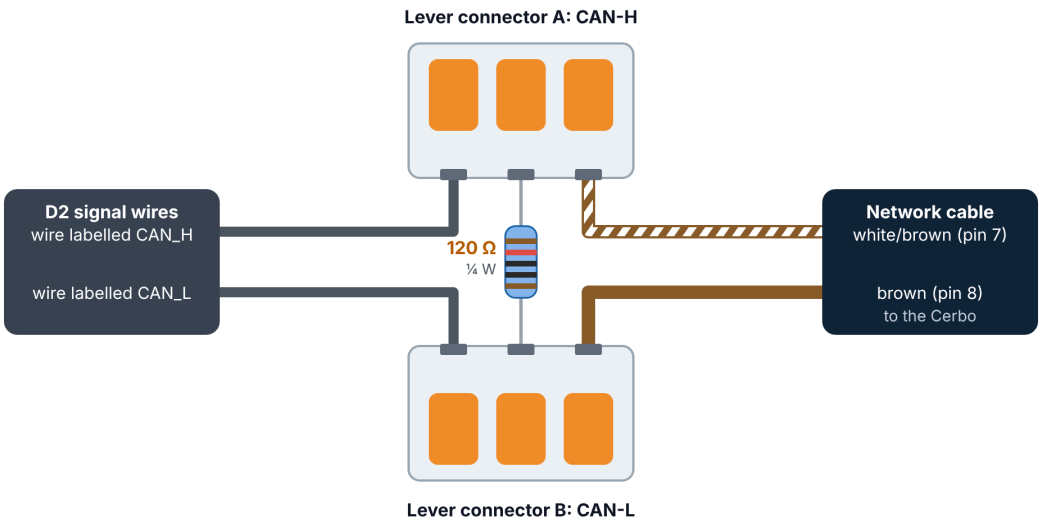


white/brown = CAN_H, brown = CAN_L. White/green and the other wires: not used.

The D2 powers its controller from its LV+ or HV+ terminal.

Blue: the network cable you make. The two lever connectors join it to the D2's CAN_H and CAN_L wires, with the required 120 Ω resistor between them. The D2 has no KL30/KL15 wires: it powers itself from its LV+/HV+ terminals.

The two lever connectors

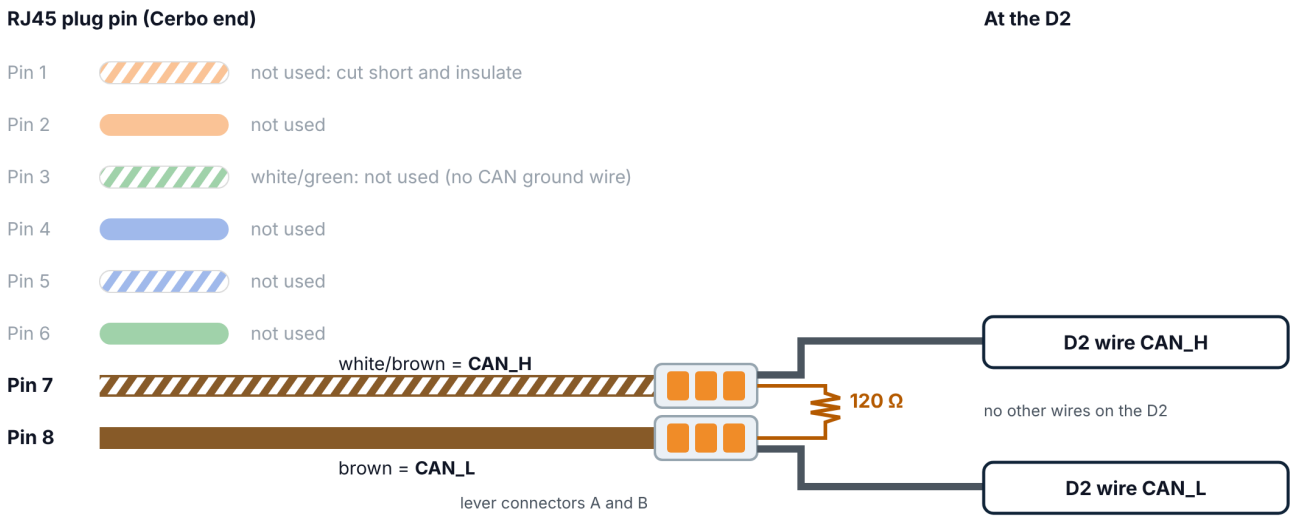


White/green (pin 3) and the other five wires: **not used**, cut short and insulate each one. No ground wire.

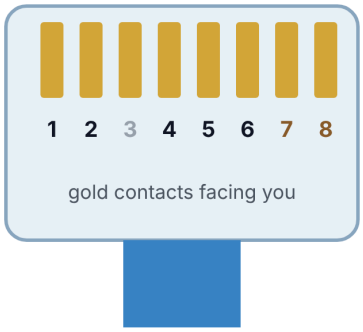
Strip each wire 11 mm (the gauge on the connector's side), push it fully in, close the lever. Heat-shrink both resistor leads.

Each connector joins three things: one D2 wire, one network-cable wire and one leg of the resistor. The resistor bridges the two connectors, so it sits across CAN_H and CAN_L at the D2.

Which wire goes where

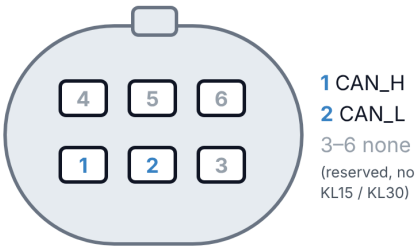


Pins 7 and 8 are the **brown pair** (twisted together), which suits CAN well. Everything else, including white/green (pin 3), is not used.



cable pointing down, clip at the back

RJ45 plug (Cerbo end): only pins 7 and 8 are used.



layout as drawn in the D2 manual (facing the D2's connector)

D2 signal plug (TE 1-967616-1), for reference only: its wires come out labelled, so there's no need to open it.

Wire	RJ45 pin (Cerbo)	Goes to
white/brown	7	**Lever connector A**, with the D2's wire labelled **CAN_H**
brown	8	**Lever connector B**, with the D2's wire labelled **CAN_L**
120 Ω resistor	–	One leg in connector A, the other in connector B
The other 6 wires (pins 1–6, including white/green)		Not used. Cut them short and insulate each one. No ground wire

Sources: Victron's *VE.Can to CAN-bus BMS cables manual* (Cerbo side: pin 7 CAN-H, pin 8 CAN-L) and the *D2 Bidirectional DC-DC Converter Manual*, Rev D.

Make it

2a **Cut off one plug.** Keep the other plug for the Cerbo. Strip about 5 cm (2 in) of the grey outer jacket.

2b **Check the wires with a multimeter** on the continuity (beep) setting. Touch one probe to a gold contact on the plug and the other to a wire:

- Pin 7 should beep with **white/brown**
- Pin 8 should beep with **brown**

2c **Insulate the 6 wires you don't need**, including white/green. Cut them short and cover **each one** separately with heat-shrink or tape, so none can touch anything.

2d **Join the wires with the two lever connectors**, with the D2 switched off. First, with everything off, measure across the D2's CAN_H and CAN_L wires: if it reads about 120 Ω the D2 has a terminator built in, so leave the resistor out. Strip each wire **11 mm** (the strip gauge on the connector's side), lift the lever, push the wire fully in, and close the lever.

- **Connector A (CAN-H):** the D2's wire labelled **CAN_H**, the **white/brown** wire, and one leg of the resistor
- **Connector B (CAN-L):** the D2's wire labelled **CAN_L**, the **brown** wire, and the other leg of the resistor

Slide heat-shrink over both resistor legs so they can't touch each other or anything metal. Keep the brown pair twisted up to a few centimetres from the connectors. Tie the connectors and cable to something solid, or put them in a small box, so road vibration can't pull on them.

2e **No ground wire.** CAN works on the difference between CAN_H and CAN_L. The Cerbo and the D2 already share the van's negative through their power wiring, so a separate CAN ground isn't needed. Leaving it off also avoids a thin wire running in parallel with the main negative cables (the Cerbo's VE.Can 2 port is not isolated). Once everything runs, step 5c includes a quick check of the ground difference.

> Never connect any of the network-cable wires to the D2's HV+ or LV+ terminals. They only join the D2's labelled CAN_H and CAN_L wires.

2f **No other wires.** Unlike the D1, the D2 has no KL30 or KL15. It powers itself from whichever of its LV+ and HV+ terminals has power, and it talks on CAN whenever it is powered. If you are replacing a D1, insulate its old KL15 and KL30 wires.

2g **Check the resistor.** With the RJ45 plug not yet in the Cerbo, measure between pins 7 and 8 of the RJ45 plug (or the white/brown and brown wires). It should read about **120 Ω**. If it reads open, a wire isn't fully in its lever connector.

2h **Final check:** plug the RJ45 into the Cerbo port you'll use, with the VE.Can terminator in that port's other socket. With everything switched off, measure between white/brown and brown again:

You measure	Meaning
About 60 Ω	Correct: both ends are terminated
About 120 Ω	The VE.Can terminator isn't in the Cerbo port's other socket, or the cable is in a different port
Very high or "OL"	The resistor at the D2 isn't connected, or a wire isn't fully in its lever connector. Re-check steps 2d and 2g

3 Set up the Cerbo

Do this on the Cerbo's touchscreen, or remotely with Remote Console in VRM. Every item below is found from the Settings menu.

3a **Check the firmware version.** Go to **Settings › General › Firmware**. It needs to be **v3.80 or newer**. If it's older, use **Online updates** on the same page to update first.
Then go to **Settings › General › Modification checks** and make sure **modifications are enabled** (that's the default). If they are off, Venus OS switches off the start-up script and the driver disappears after the next restart.

3b **Write down the Cerbo's IP address.** Go to **Settings › Connectivity › Ethernet** (or Wi-Fi) and read the IP address.
IP address: _____

3c **Choose the CAN port and speed.** Use the Cerbo's second CAN port, with nothing else plugged into it. The labels next to the ports tell you which Cerbo you have:

- **Cerbo GX MK2** (ports labeled **VE.Can 1** and **VE.Can 2**): go to **Settings › Connectivity › VE.Can port 2** (some firmware still calls it **BMS-Can port**) and set **CAN-bus profile** to **VE.Can (250 kbit/s)**. That is the default. The D2 talks at either 250 or 500 kbit/s, so start with 250. Step 5c tells you if the D2 is talking at the other speed. If it is, change the profile to **CAN-bus BMS LV (500 kbit/s)**. Older firmware calls it **CAN-bus BMS (500 kbit/s)**.
  Victron's own messages use a different kind of CAN address from the D2's, so they don't get in each other's way. Other guides may mention a profile called *Up, but no services*. The Cerbo screen no longer offers it, and you don't need it.
- **Original Cerbo GX** (second port labeled **BMS-Can**): go to **Settings › Connectivity › BMS-Can port**. This port only runs at 500 kbit/s, so leave it on **CAN-bus BMS (500 kbit/s)**. It works only if the D2 talks at 500 kbit/s. If step 5c reports bus errors on it, the D2 is at 250 kbit/s: use the VE.Can port (if nothing else is on it) with **VE.Can (250 kbit/s)** instead.

3d **Let your PC connect to the Cerbo** (this switches on SSH):

1. Go to **Settings › General › Access & Security › Access level** and choose **User & Installer**. The password is `ZZZ`.
2. Still on the Access & Security page, drag the list down with your finger and hold it for about 5 seconds (on Remote Console with a keyboard: hold the right-arrow key). The access level changes to **Superuser**.
3. Tap **Root password** and choose a password (at least 8 characters).
   Root password: _____
4. Switch **Enable SSH on LAN** to on.

4 Connect everything

4a Plug the cable's RJ45 plug into the CAN port you chose in 3c, with the VE.Can terminator in the port's other socket (as in step 2h).

4b Plug the D2's signal plug back in and close the fuses or breakers on its LV+ and HV+ leads.

> Nothing changes for the D2, now or later. The monitor can only listen: it has no way to send the D2 a command.

5 Install the monitor with Cerbo Setup

5a **Open the helper.** In the project folder, double-click `Cerbo Setup.bat`. If Windows shows "Windows protected your PC", click **More info** and then **Run anyway**.

5b **Type the Cerbo's IP address** from step 3b and press **Enter**. The helper remembers it for next time, and then shows its menu:

```
  ============================================================
    Nomadic DC-DC monitor  -  Cerbo GX setup
    Cerbo address: 192.168.1.50
    Monitor only: nothing is ever sent to the converter
  ============================================================

    1  Edit the settings (config.ini)
    2  Find which Cerbo port the converter is on
    3  Watch live readings from the converter
    4  Install or update the monitor on the Cerbo
    5  Show the monitor log

    U  Make a USB-stick installer (no network needed)

    9  Try it with a pretend converter (simulation)

    6  Change the Cerbo address
    7  Fix "REMOTE HOST IDENTIFICATION HAS CHANGED"
    8  Uninstall the monitor from the Cerbo
    0  Exit

  Type a number and press Enter: _
```

> **Want to see it first?** Option **9** runs everything with a pretend D2 and no CAN bus at all (see the Safe Test Plan, `docs/safe-test-plan.pdf`).

> **The password prompt.** Every option that talks to the Cerbo asks for the root password from step 3d. **Nothing appears on screen while you type it.** That's normal: type it and press Enter.

5c **Type `2`: find the converter.** The helper copies the monitor to the Cerbo and listens on every CAN port for 3 seconds. You should see something like this:

```
Venus OS v3.80 OK
Listening 3 s on each CAN port (nothing is transmitted)...
  can0    250 kbit/s  error-active  nothing from the converter
  can1    250 kbit/s  error-active  CONVERTER FOUND (391 frames)
The converter is on can1. config.ini has interface = auto, so the driver finds it there by itself.
Good.
```

Port names differ between Cerbo models (`can1`, or `vecan1` on the MK2). That's fine: with `interface = auto` the driver uses whichever port the D2 is on. If the D2 is set to the other speed, you'll see bus errors

instead:

```
  can1    250 kbit/s  error-passive nothing decoded, but 288 bus errors: talking at a different
speed?
Something is talking on can1 at a different speed. On the Cerbo, set that port's CAN-bus
profile to "CAN-bus BMS LV (500 kbit/s)" and run this again.
```

Do what it says (step 3c), then type `2` again. If nothing is heard at all, see Troubleshooting, and check first that the D2 has power on its LV+ or HV+ terminal.

> **Ground check (no CAN ground wire).** Later, with the D2 working hard, measure between the Cerbo's negative terminal and the D2's GND terminal. A few tenths of a volt is normal. More than about 1 V points to a poor negative connection: fix that first. If the Cerbo ever can't hear the D2 although the wiring and the 60 Ω check are right, try a wire from RJ45 pin 3 (white/green) to the D2's GND as a test.

5d **Type `1` : edit the settings.** Notepad opens `config.ini`. Check these lines, then press Ctrl+S to save and close Notepad:

Line	What to put
`interface = auto`	Leave as auto: the driver finds the D2's CAN port by itself
`custom_name = DC-DC D2`	The name shown on the Cerbo and in VRM
`password =`	Optional: a password for the web page. Without one, anyone on your network can see the readings (but can't change anything)
`switch_pane = yes`	Leave as yes to get the cards on the Cerbo screen and in VRM

5e **Type `3` : watch the live readings** (optional, but a good check). The voltages should match what a meter shows on the D2's terminals. To stop, press Ctrl+C, then type N and press Enter.

```
--- can1 10:42:07  ONLINE  0x066=20/s 0x088=100/s 0x520=10/s
LV  13.4 V  +22 A   +295 W | HV  51.2 V   -6 A   -307 W | 35 C
Mode: Autonomous (default set-point) | Precharge not in process
Alarms: none
Software 1.2.3  Hardware 5
```

5f **Type `4` : install.** After about 10 seconds you should see:

```
/service/dbus-nomadic-dcdc: up (pid 1834) 6 seconds
... INFO: Nomadic DC-DC monitor 2.4.0 starting on auto (monitor only: nothing is transmitted)
... INFO: Converter online on can1
```

Done! It now starts by itself after every reboot and firmware update.

No network to the Cerbo? Install from a USB stick instead

Venus OS can install add-ons from a USB stick or microSD card by itself, with no SSH or password. Do this instead of 5c–5f:

U1 **Get the file.** Download `venus-data-nomadic-d2-monitor.tar.gz` from the latest release (https://github.com/CKOCI13/CKOCI13-cerbo-nomadic-d2-monitor/releases/latest). It has the standard settings, with `interface = auto`, so there's nothing to edit. To change settings first, edit config.ini (Cerbo Setup option `1`) and type `U` to make your own copy of the file. On a Mac, download it with Chrome or Firefox: Safari can unpack it into a `.tar` file, which the Cerbo ignores.

U2 Copy it to the **top folder** of a USB stick or SD card (FAT32), not inside another folder. Put only **one** `venus-data` file on the stick. If your browser saved it as `venus-data-nomadic-d2-monitor (1).tar.gz`, rename it back to `venus-data-nomadic-d2-monitor.tar.gz`: the Cerbo skips names with spaces.

U3 Put the stick in the Cerbo and restart it (**Settings › General › Reboot**, or switch it off and on). The Cerbo unpacks the file and installs the monitor while it starts.

U4 **Wait about 2 minutes, then take the stick out.** The Cerbo renames the file on the stick to `...tar.gz.installed`, so it never installs twice. Then open the switch pane: the DC-DC D2 cards should be there.

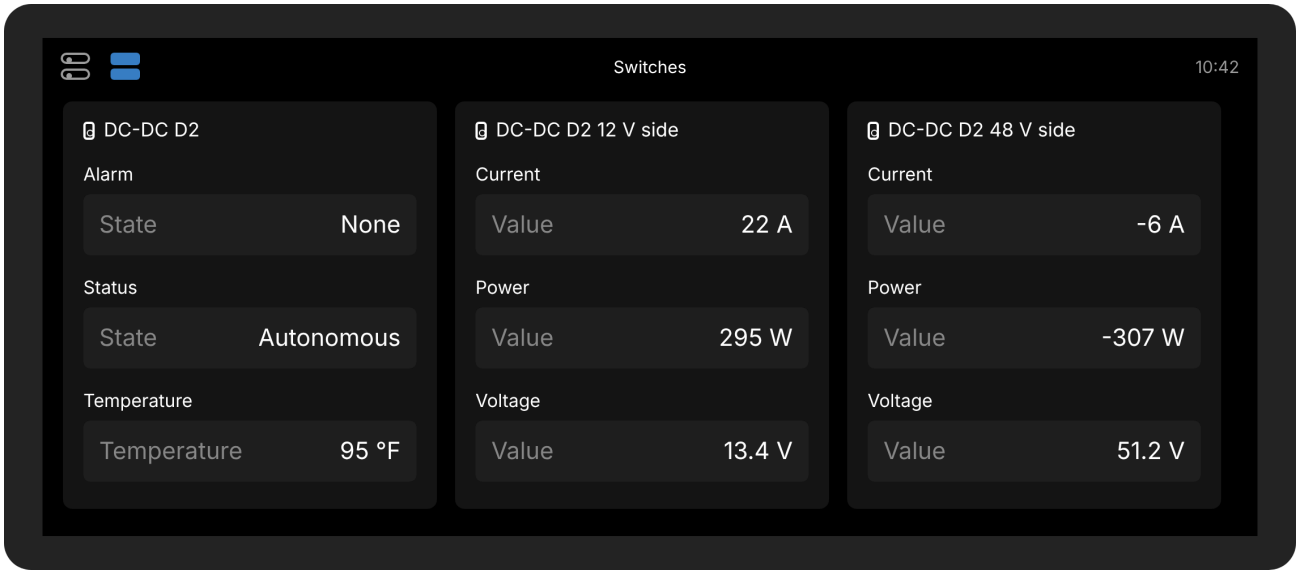
> Without SSH you can't run "find the converter" (5c). If the Status card says **No communication**, check the CAN port you chose in 3c, its speed (250 / 500 kbit/s) and the wiring, and restart the Cerbo. To install again, use a fresh file (U1).

> **Upgrading from the old control version (1.x)?** Installing this monitor (option 4) replaces it. If the old version was in Off, Buck or Boost, its commands stop when the monitor starts. Nobody knows yet whether the D2 then goes back to its own logic or keeps the last command. So first switch the old version to **Automatic** (it then sends nothing), check the D2 behaves as it does on its own, and only then install. If it seems stuck, restart its controller by opening the fuses or breakers on both LV+ and HV+ (it has no KL15).

6 Use it

On the Cerbo screen (and Remote Console)

Tap the **switches icon** at the top left of the screen. Three cards appear (swipe if your screen shows fewer at once):



Card	Item	Shows
DC-DC D2	Alarm	The most serious active alarm, or None (the D2's own alarms first, then the monitor's temperature warning). The tile turns red on a fault or a temperature alarm, and "N active" appears when there are several
	Status	What the D2 says it's doing: Standby, Autonomous, Buck, Boost, Derating (hot)… or **No communication**
	Temperature	The D2's power board temperature, in your Cerbo's temperature unit. For °F choose Fahrenheit at **Settings › General › Display & Appearance › Data units › Temperature**
DC-DC D2 12 V side **DC-DC D2 48 V side**	Current, Power, Voltage	Live readings for that battery side. **+** means energy going into that battery, **–** coming out (see below)

To rename, move or hide any item, go to **Settings › Devices › DC-DC D2**. A hidden **Precharge status** item is there too (the D2 manual doesn't use precharge).

On the Overview page

The D2 has an **Alternator** tile: **+** watts while it charges the 48 V battery, **–** while it powers the 12 V side from it. Listing it as an alternator keeps the Cerbo's **DC Loads** figure right. Nothing is sent to the D2. For no tile, set `service_class = dcdc`.

The D2's temperature, on every screen

The temperature is the D2's **internal power board**, the same sensor the D2 uses to protect itself. It runs well above the air around it: a D1 read about 180 °F while boosting about 1.7 kW, which is normal.

Where	What you see
Overview page	Top right of the **Alternator** tile
Device page (Settings › Devices › DC-DC D2)	**Alternator temperature** (Victron's fixed wording)
Levels › Environment	A gauge named **DC-DC D2**, 32–212 °F. Its sensor type is **Water heater**, the only type whose scale covers the D2's working range
Brief page	Can show it in the centre circle: Settings › General › Display & Appearance › Brief page › Center details
Relays	A Cerbo relay set to the **Temperature** function can switch a fan or light on its reading (Settings › Relay › Temperature control rules)
Switch pane, web page, VRM	Temperature item; the web page also says OK / Warning / Alarm

Board temperature	Converter (D1 manual Rev D, IV.B; the D2 manual gives the same 248 °F stop)	Monitor
up to 212 °F (100 °C)	Full output	OK
212–221 °F (100–105 °C)	Full output	**Warning**: High Temperature notification, Alarm card "Temperature high"
221–248 °F (105–120 °C)	Cuts its current back, from 100 % to 0 %; status "Derating (hot)"	**Alarm**: notification, red Alarm card
above 248 °F (120 °C)	Stops and reports a fault	The D2's own fault on the Alarm card

The levels are in the `[temperature]` part of config.ini, in °C. Each clears 5 °F (3 °C) below where it came on. The manual asks for airflow along the heatsink fins while it works.

In the VRM app and portal

The same cards appear in the **Switches** widget in the VRM app, and in the **Switches** panel on the VRM portal dashboard. The Cerbo must be online with VRM. Nothing needs to be changed in VRM's access settings: there is nothing to control.

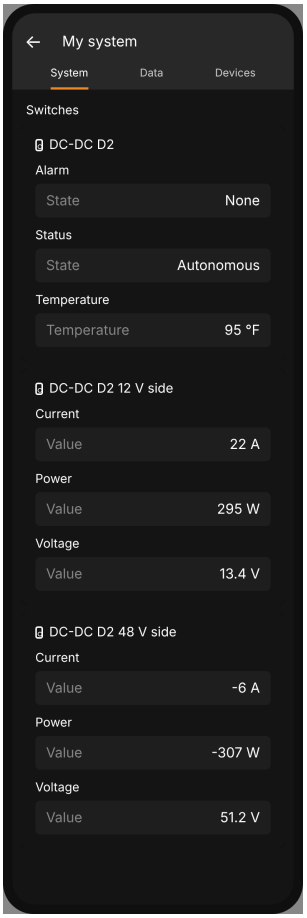
The picture is an illustration. VRM draws its own version of these cards, so the layout may differ a little.

On a web page, with charts

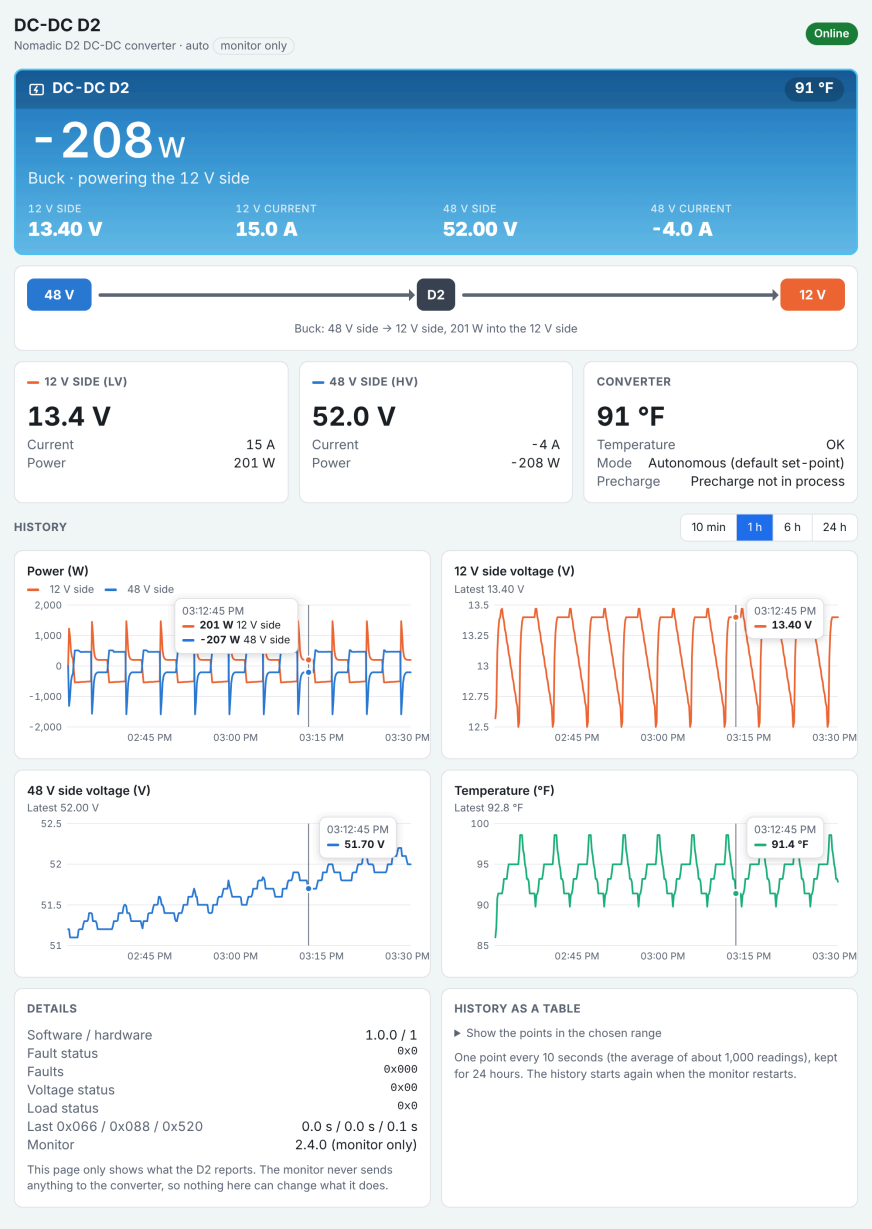
On any phone, tablet or computer on the same network, open:

http://<Cerbo IP>:8088

It shows the live readings, which way the energy is flowing, every alarm, the raw status bits, and charts of the last 10 minutes to 24 hours. Point at a chart to read the values at that moment. The history starts again when the monitor restarts.

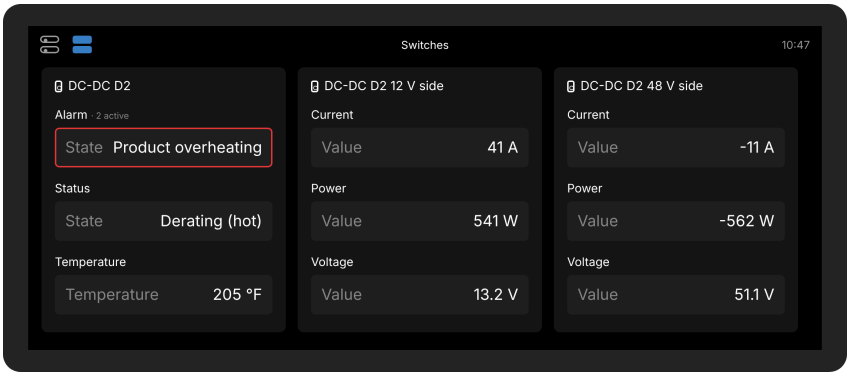


VRM app, Switches widget (illustration)



The web page, with a pretend D2 (simulation).

When something is wrong



An example: the D2 is hot, so it's limiting its power (derating). The alarm shows the most serious problem first.

Reading the numbers

Which way is the energy going?

BUCK: 48 V HELPS 12 V

48 V ····· D2 ····· 12 V

12 V side current **+22 A** · power **+295 W**
48 V side current **−6 A** · power **−307 W**

BOOST: 12 V HELPS 48 V

48 V ····· D2 ····· 12 V

12 V side current **−40 A** · power **−520 W**
48 V side current **+9 A** · power **+470 W**

+ means energy is going **into** that battery, **−** means it's coming **out**. The side that's being charged is the one with the plus sign. The 48 V side always shows a little more than the 12 V side gets in buck (and the other way round in boost): that difference is the D2's own losses.

The D2's documents don't say which way its current is counted. The monitor shows exactly what the D2 reports; check the sign once with a clamp meter.

On its own, the D2 **bucks** (charges the 12 V side from 48 V) when the 12 V side drops below 12.5 V. It only does this when the 48 V side is 41–58 V, and it fills the 12 V side back up to about 13.4 V. It **boosts** (pushes 12 V energy up to 48 V) when the 12 V side is 13.5–16 V, until it falls below 12.3 V. In between it waits in **Standby**, with no current.

? Troubleshooting

What you see	What to do
Option 2: "not heard on any port"	Check, in this order: 1. the D2 has power: its LV+ terminal (9–16 V) or HV+ terminal (41–58 V), measured to GND. With neither, its controller is off and silent 2. the CAN speed: if it reports bus errors, switch to the other speed (step 3c) 3. the wires (2b), and that each lever connector holds the right D2 wire (CAN_H with white/brown, CAN_L with brown) 4. the termination reads about 60 Ω (2h) 5. the cable is in the Cerbo port you set up
"ssh" or "tar" is not recognized	Install the Windows OpenSSH client: Settings › Apps › Optional features › Add a feature › OpenSSH Client.
"Connection timed out" or "Connection refused"	Check the IP address (menu option 6), that the PC and Cerbo are on the same network, and that Enable SSH on LAN is on (step 3d).
"Permission denied"	Wrong root password. Set it again (step 3d, item 3).
"REMOTE HOST IDENTIFICATION HAS CHANGED"	This happens after the Cerbo is reset or replaced. Use menu option 7, then try again.
Option 2: Venus OS version warning	Update the Cerbo (step 3a). Older firmware can't show the cards.
Nothing appears in the switch pane	Check the firmware is v3.80 or newer. Use option 5 (log) and look for "Registered ... on D-Bus". Make sure `switch_pane = yes` in config.ini.
Status shows Standby	Normal when the 12 V side is between 12.5 and 13.5 V: the D2 has nothing to do.
Status shows No communication	The Cerbo can't hear the D2. Check that its LV+ or HV+ terminal is live, the cable and the termination.
The charts say "Collecting data..."	They add a point every 10 seconds. Wait a minute. They start empty after every restart of the monitor or the Cerbo.
The "DC Loads" figure on the Cerbo goes negative	The D2 is set up as a DC-DC converter (`service_class = dcdc`), which Victron leaves out of its sums. Set `service_class = alternator` (the default) in config.ini (option 1, then 4).

To change settings later, open Cerbo Setup, choose 1 to edit config.ini, save, then choose 4. **To remove it,** choose 8 and type YES. The D2 is not affected either way.